

HW 10

DATA SET

DS1406

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.⁹ Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

Educational Level	Married		
	Yes	No	
Bachelors or higher	187	113	300
Some college	162	138	300
High school or less	149	151	300

498 402

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

① $H_0 =$ 婚姻率与教育程度无关

$H_a =$ 有关

② $\alpha = 0.01$

③ $df = 2, 1 = 2$

④ $E_{11} = \frac{300 \cdot 498}{900} = 166$

$E_{12} = 166$

$E_{13} = 166$

$E_{21} = \frac{402 \cdot 300}{900} = 134$

$E_{22} = 134$

$E_{23} = 134$

$$\chi^2 = \frac{(187-166)^2}{166} + \frac{(162-166)^2}{166}$$

$$+ \frac{(149-166)^2}{166} + \frac{(113-134)^2}{134} + \frac{(138-134)^2}{134}$$

$$+ \frac{(151-134)^2}{134} = 10.06$$

$$\textcircled{5} \chi^2_{0.01, 2} = 9.21$$

$\textcircled{6} \text{ rej } H_0 \#$

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

Family Members	Type of Residence		
	Apartment	Duplex	Single Residence
1	8	20	1
2	16	8	9
3	10	10	14
4 or More	6	2	16

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

①

$H_0 =$ 三種分布相同

$H_a =$ 有一個不同

② $\alpha = 0.01$ ③ $df = 3 - 1 = 2$ ④ $\chi^2 \approx 28.17$ ⑤ $\chi^2_{0.01, 2} = 9.210$ ⑥ $\text{rej } H_0$

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

- Use a one-tailed (Wilcoxon rank sum test) with $\alpha = .05$ to confirm or refute the theory.
- Use a Student's t -test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

* 不是 paired, 不算
差值, rank sum

(a) ① $H_0 = \text{Above} = \text{below}$
 $H_a = \text{Above} > \text{below}$

② $\alpha = 0.05$

③

4.7	B	1
4.8	A	2.5
4.8	B	2.5
4.9	A	5
4.9	B	5
4.9	B	5
5	A	7.5
5	B	7.5
5.1	A	9
5.2	A	10

$$\frac{4+5+6}{3} = 5$$

$$T_1 = 2.5 + 5 + 7.5 + 9 + 10 = 34$$

$$T_1^* = 5(11) - 34 = 21$$

$$T_{0.05, 5} = 19$$

④ $21 > 19$, $\text{rej} = T \leq 19$

⑤ Don't rej H_0 #

(b)

$$\textcircled{3} \quad \bar{x}_1 = \frac{4.8 + 5.2 + 5 + 4.9 + 5.1}{5} = 5$$
$$\bar{x}_2 = 4.86$$

$$\bar{x}_1 - \bar{x}_2 = 0.14, \quad s_1^2 = 0.025, \quad s_2^2 = 0.019$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = 0.019$$

$$s_p = 0.138$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = 1.61, \quad df = n_1 + n_2 - 2 = 8$$

④ $t_{0.05, 8} = 1.86$

⑤ $1.61 < 1.86$, Don't rej H_0 #

0.05

■ Table 7 Critical Values of T for the Wilcoxon Rank Sum Test, $n_1 \leq n_2$

■ Table 7(a)
5% Left-Tailed
Critical Values

n_2	n_1													
	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	—	6												
4	—	6	11											
5	3	7	12	19										
6	3	8	13	20	28									
7	3	8	14	21	29	39								
8	4	9	15	23	31	41	51							
9	4	10	16	24	33	43	54	66						
10	4	10	17	26	35	45	56	69	82					
11	4	11	18	27	37	47	59	72	86	100				
12	5	11	19	28	38	49	62	75	89	104	120			
13	5	12	20	30	40	52	64	78	92	108	125	142		
14	6	13	21	31	42	54	67	81	96	112	129	147	166	
15	6	13	22	33	44	56	69	84	99	116	133	152	171	192

■ Table 7(b)
2.5% Left-Tailed
Critical Values

n_2	n_1													
	2	3	4	5	6	7	8	9	10	11	12	13	14	15
4	—	—	10											
5	—	6	11	17										
6	—	7	12	18	26									
7	—	7	13	20	27	36								
8	3	8	14	21	29	38	49							
9	3	8	14	22	31	40	51	62						
10	3	9	15	23	32	42	53	65	78					
11	3	9	16	24	34	44	55	68	81	96				
12	4	10	17	26	35	46	58	71	84	99	115			
13	4	10	18	27	37	48	60	73	88	103	119	136		
14	4	11	19	28	38	50	62	76	91	106	123	141	160	
15	4	11	20	29	40	52	65	79	94	110	127	145	164	184

Source: Data from "An Extended Table of Critical Values for the Mann-Whitney (Wilcoxon) Two-Sample Statistic" by Roy C. Milton, pp. 925–934 in the *Journal of the American Statistical Association*, Volume 59, No. 307, Sept. 1964. Reprinted from the *Journal of the American Statistical Association*.

■ Table 7(c)
1% Left-Tailed
Critical Values

n_2	n_1													
	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	—	—												
4	—	—	—											
5	—	—	10	16										
6	—	—	11	17	24									
7	—	6	11	18	25	34								
8	—	6	12	19	27	35	45							
9	—	7	13	20	28	37	47	59						
10	—	7	13	21	29	39	49	61	74					
11	—	7	14	22	30	40	51	63	77	91				
12	—	8	15	23	32	42	53	66	79	94	109			
13	3	8	15	24	33	44	56	68	82	97	113	130		
14	3	8	16	25	34	45	58	71	85	100	116	134	152	
15	3	9	17	26	36	47	60	73	88	103	120	138	156	176

■ Table 7(d)
.5% Left-Tailed
Critical Values

	n_1													
n_2	3	4	5	6	7	8	9	10	11	12	13	14	15	
3	—													
4	—	—												
5	—	—	15											
6	—	10	16	23										
7	—	10	16	24	32									
8	—	11	17	25	34	42								
9	6	11	18	26	35	45	56							
10	6	12	19	27	37	47	58	71						
11	6	12	20	28	38	49	61	73	87					
12	7	13	21	30	40	51	63	76	90	105				
13	7	13	22	31	41	53	65	79	93	109	125			
14	7	14	22	32	43	54	67	81	96	112	129	147		
15	8	15	23	33	44	56	69	84	99	115	133	151	171	

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

unpaired

	Deaf Children	Hearing Children
	2.75 (15)	.89 (1)
	2.14 (11)	1.43 (7)
	3.23 (18)	1.06 (4)
	2.07 (10)	1.01 (3)
	2.49 (14)	.94 (2)
	2.18 (12)	1.79 (8)
	3.16 (17)	1.12 (5.5)
	2.93 (16)	2.01 (9)
	2.20 (13)	1.12 (5.5)
Rank Sum	126	45

① $H_0 = \text{same}$

$H_a = \text{differ}$

② $\alpha = 0.05$

③ $T_1 = 126, T_1^* = 9(19) - 126 = 45$
 $\min(T_1, T_1^*) = 45$

④ $T_{0.025, 9} = 62$

⑤ $\text{rej} = T \leq 62, 45 < 62$

⑥ $\text{rej } H_0$

Two Simple Examples Use the (sign test) to compare two populations for significant differences for the paired data in Exercises 4-5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \leq .10$. Calculate the observed value of the test statistic and present your conclusion.

4.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

+ + + - + + +

① $H_0 = p = 0.5$

$H_a = p \neq 0.5$

② $\alpha = 0.1$

✓

③ $x = 6$ $P(X=0) = 0.1008$

$n = 7$

< 0.05

可放入 #

拒绝 H₀

$P(X \leq 1) = 0.1062 > 0.05$

不放入

$rej = x \leq 0, x \geq 7$

∴ 左尾只可能 $x=0$ (全-)

右 " $x=7$ (全+)

但算出 $x=6$, 因此 don't rej H_0 #

$n = 7$ [illegible]

11. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1–10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

Meal	A	B	Meal	A	B
1	6	8	12	8	5
2	4	5	13	4	2
3	7	4	14	3	3
4	8	7	15	6	8
5	2	3	16	9	10
6	7	4	17	9	8
7	9	9	18	4	6
8	7	8	19	4	3
9	2	5	20	5	4
10	4	3	21	3	2
11	6	9	22	5	3

- Use the binomial tables in Appendix I to find the exact rejection region for the test.
- Use the large-sample z statistic. (NOTE: Although the large-sample approximation is suggested for $n \geq 25$, it works fairly well for values of n as small as 15.)
- Compare the results of parts a and b.

a. ① $H_0 = p = 0.5$

$H_a = p \neq 0.5$

② $\alpha = 0.05$

③

$X = 11, n = 22 - 2 = 20$

$\alpha/2 = 0.025$

$P(X \leq 5) = 0.021 \quad \checkmark$

$P(X \leq 6) = 0.058$

④

⑤

rej = $X \leq 5, X \geq 15$, Don't rej

H_0 #

$n = 20$ [illegible]

13. Recovery Rates Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.

- a. Test using the (sign test.) Choose your rejection region so that α is near .05.
- b. Why might it be inappropriate to use the Student's t -test in analyzing the data?

Drug A			
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	84	63	75.0
2	63	44	69.8
3	56	48	85.7
4	77	57	74.0
5	29	20	69.0
6	48	40	83.3
7	61	42	68.9
8	45	35	77.8
9	79	57	72.2
10	62	48	77.4

Drug B			
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4
2	83	69	83.1
3	91	73	80.2
4	47	35	74.5
5	60	42	70.0
6	27	22	81.5
7	69	52	75.4
8	72	57	79.2
9	89	76	85.4
10	46	37	80.4

91

① $H_0 = p = 0.5$

$H_a = p \neq 0.5$

② $\alpha = 0.05$

③

$X = 2, n = 10$

$P(X \leq 1) = 0.011 \checkmark$

$P(X \leq 2) = 0.055$

④ $rej = X \leq 1, X \geq 9$

⑤ don't rej H_0 #

$n = 10$

k	p													k
	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99	
0	.904	.599	.349	.107	.028	.006	.001	.000	.000	.000	.000	.000	.000	0
1	.996	.914	.736	.376	.149	.046	.011	.002	.000	.000	.000	.000	.000	1
2	1.000	.988	.930	.678	.383	.167	.055	.012	.002	.000	.000	.000	.000	2
3	1.000	.999	.987	.879	.650	.382	.172	.055	.011	.001	.000	.000	.000	3
4	1.000	1.000	.998	.967	.850	.633	.377	.166	.047	.006	.000	.000	.000	4
5	1.000	1.000	1.000	.994	.953	.834	.623	.367	.150	.033	.002	.000	.000	5
6	1.000	1.000	1.000	.999	.989	.945	.828	.618	.350	.121	.013	.001	.000	6
7	1.000	1.000	1.000	1.000	.998	.988	.945	.833	.617	.322	.070	.012	.000	7
8	1.000	1.000	1.000	1.000	1.000	.998	.989	.954	.851	.624	.264	.086	.004	8
9	1.000	1.000	1.000	1.000	1.000	1.000	.999	.994	.972	.893	.651	.401	.096	9
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	10

6、

因這些資料是百分比，不是一般連續變測量值，人又不一樣，百分比不穩定度不同